Fig. 4. Controlled case N6: a source-level uncovered cleanup path compiled to a single-EXIT binary via predication

Source (CUDA)

```
if (tid < 16) return;   // 16 of 32 threads exit HERE
// -- cleanup below is source-level UNCOVERED for tid<16 -
volatile uint32_t *v = sbuf;
for (...) v[i] = 0u;      // zero-store
__threadfence_block();     // fence
__syncthreads();           // barrier
```

ptxas (predication) →

Compiled SASS (both toolkits, all 8 configs)

```
...predicated skip for tid<16 (no separate EXIT)...
STS [...], RZ     <- zero-store  @ line 166
MEMBAR.SC.CTA     <- fence        @ line 174
BAR.SYNC          <- barrier      @ line 176
EXIT              <- ONLY exit, count=1
```

Checker verdict: FULL_SIGNATURE (cross-tool agreement: True)

Interpretation: the checker correctly reports the recognizable static signature present before the single EXIT instruction in the disassembly. It does not, and by its documented scope cannot, determine that the 16 threads which returned early never reached this sequence.

This is direct, compiled, cross-tool-agreed evidence that a FULL_SIGNATURE verdict establishes static structural presence and ordering -- not path-complete dynamic execution.